



Hardware-Aware Low-Light Image Enhancement for Improved Visual Quality

Praneel Vipul Vania (202201131)
202201131@dau.ac.in

Aditya Sable (202201049)
202201049@dau.ac.in

Dhirubhai Ambani University (DAU)

Course Code: PC433

Under the guidance of: Prof. **Pankaj Kumar**

REPORT

Abstract—Images captured in low-light conditions often exhibit reduced visibility due to insufficient brightness, poor contrast, noise, and loss of structural details. Enhancing such images while maintaining their natural appearance is a significant challenge in image processing. This project aims to improve visual quality by designing and analyzing an effective enhancement pipeline.

The work involves implementing an existing research-based approach, followed by a detailed evaluation of its performance and limitations. Based on these observations, several modifications and improvements are explored to achieve more stable and visually consistent results. The overall study focuses on developing a robust enhancement framework through systematic experimentation and analysis.

I. ORGANIZATION OF THE REPORT

The report is organized into three main parts:

Part 1: This part deals with understanding and implementing the original research methodology, followed by its critical analysis and proposed improvements.

Part 2: This part focuses on evaluating the approach using datasets and comparing it with other image enhancement techniques.

Part 3: This part explores more advanced enhancement pipelines and examines their practical applications.

Each part is structured to build upon the insights and results obtained in the preceding sections.

II. INTRODUCTION

Improving the quality of images captured in low-light environments is a fundamental challenge in image processing. Images taken under such conditions often suffer from low brightness, poor contrast, visible noise, and loss of important structural details. The objective of this project is to enhance these images in a way that improves visibility while preserving their natural appearance and structural consistency.

The work carried out in this phase of the BTP involves the following:

- Analyzing and reproducing an existing research-based method
- Designing and implementing the complete enhancement pipeline
- Examining the limitations of the implemented approach
- Proposing improvements and validating them through experimentation

III. STUDY AND UNDERSTANDING OF THE RESEARCH PAPER

The project began with a detailed examination of a research approach that combines a reflection-based image model with wavelet-based fusion techniques, as presented in [1].

A. Illumination–Reflection Model

The model represents an image as a combination of two key components:

- Illumination, which captures the lighting conditions of the scene
- Reflectance, which represents the inherent details of the objects in the scene

B. HSV Color Space

- The image is converted from RGB to HSV in order to separate brightness from color information
- The Value (V) channel corresponds to illumination
- The Hue (H) and Saturation (S) channels are preserved to avoid color distortion

C. Multiscale Illumination Estimation

- Illumination is approximated using Gaussian filters applied at multiple scales to capture both local and global brightness variations

D. Adaptive Brightness Enhancement

- The enhancement approach is inspired by human visual perception, particularly the Weber–Fechner law
- The strength of enhancement is dynamically adjusted based on the statistical characteristics of the image

E. Wavelet-Based Fusion

- Multiple enhanced versions of the image are combined using wavelet-based fusion
- This approach aims to achieve a balance between improved brightness and preservation of fine details

IV. IMPLEMENTATION OF THE ORIGINAL PIPELINE

The enhancement pipeline proposed in the research paper was implemented in a systematic manner using Python.

A. Input Processing and Color Space Conversion

- The input image was initially loaded using OpenCV
- It was then converted from BGR to RGB, followed by transformation into the HSV color space

The HSV channels were separated as:

- $H \rightarrow$ represents color information
- $S \rightarrow$ represents saturation
- $V \rightarrow$ represents brightness

Observation: The V channel was found to effectively represent illumination, confirming that enhancement operations should primarily be applied to this component.

B. Illumination Estimation using Gaussian Filtering

Illumination was estimated using Gaussian filtering at multiple scales:

- Single-scale filtering with ($\sigma = 15$)
- Multiscale filtering using:
 - $\sigma = 15$
 - $\sigma = 80$
 - $\sigma = 250$

The final illumination estimate was computed as:

$$I = \frac{I_{15} + I_{80} + I_{250}}{3} \quad (1)$$

Observation:

- Multiscale filtering captured overall illumination trends effectively
- However, important edge details were blurred during the process
- The estimated illumination remained coarse and lacked fine precision

C. Adaptive Brightness Enhancement

The V channel was enhanced using an adaptive strategy:

- A parameter c was calculated based on:
 - Mean value of the V channel
 - Median value of the V channel
 - Saturation information

Two enhanced outputs were generated:

- V_1 corresponding to moderate enhancement
- V_2 corresponding to stronger enhancement

Observation:

- V_1 preserved structural details but offered limited improvement in brightness
- V_2 enhanced visibility more effectively but often resulted in over-bright or unnatural appearance

D. Wavelet-Based Fusion

To combine the advantages of both enhanced outputs:

- Discrete Wavelet Transform (DWT) was applied
- The image was decomposed into:
 - Approximation coefficients (LL)
 - Detail coefficients (LH, HL, HH)

Fusion was performed using:

- Maximum coefficient selection

The enhanced V channel was reconstructed using inverse DWT.

Observation:

- The fusion step successfully combined features from both enhanced images
- However, it introduced abrupt transitions in certain regions
- Noise amplification was observed, particularly in darker areas
- The output did not always maintain a natural appearance

E. Final Image Reconstruction

- The fused V channel was combined with the original H and S channels
- The image was converted back to the RGB color space

Result:

- A clear improvement in overall brightness was achieved
- Over-enhancement was observed in brighter regions
- Some degree of color distortion affected visual naturalness
- Artifacts were noticeable in certain parts of the image

V. DEBUGGING AND VALIDATION

During the implementation process, several technical challenges were encountered and addressed.

Issues and Fixes

- Dimension mismatches after wavelet reconstruction were resolved by appropriately cropping the output
- Data type inconsistencies were handled by converting values to `float32`
- Normalization issues were corrected using clipping within the range of 0–255

Intermediate Verification

- Individual H, S, and V channels were visualized for validation
- Illumination estimation results were examined at different stages
- Outputs were verified after each processing step to ensure correctness

VI. OBSERVED LIMITATIONS

The implemented pipeline exhibited several limitations:

- Blurring of edges due to Gaussian-based illumination estimation
- Excessive brightness in regions that were already well-lit
- Amplification of noise in darker areas
- Sharp or unnatural transitions caused by the max-based fusion strategy
- Inconsistent performance across different input images

VII. MODIFIED PIPELINE AND EXPERIMENTS

Based on the observed limitations, multiple modifications and experiments were carried out.

A. Guided Filter Replacement

- Gaussian filtering was replaced with guided filtering to improve edge preservation
- While edges were better maintained, the overall improvement in final output quality was limited

B. Fusion Strategy Modification

- A variance-based fusion approach was introduced
- This resulted in slightly smoother transitions compared to max fusion
- However, the overall improvement remained marginal

C. Additional Experiments

- CLAHE was applied but showed minimal impact on enhancement quality
- Unsharp masking introduced slight sharpening but did not significantly improve overall results

D. Core Insight

It was observed that the main limitation of the pipeline lies in uncontrolled brightness amplification rather than inaccuracies in illumination estimation.

VIII. RESULT



Fig. 1. Result of Original Enhancement Pipeline

IX. KEY LEARNINGS

- Processing in the HSV color space helps prevent color distortion
- The V channel serves as the appropriate component for brightness enhancement
- Illumination estimation methods have inherent limitations
- The choice of fusion strategy significantly influences output quality
- Controlling brightness is essential for achieving natural-looking results

X. CONCLUSION

The original enhancement pipeline was successfully implemented and critically analyzed. Although it effectively improves brightness, it also introduces artifacts and inconsistencies in certain cases. This phase establishes a solid foundation for further improvements and future development.

XI. PART 2: DATASET-BASED EVALUATION AND COMPARATIVE ANALYSIS

A. Motivation After Initial Implementation

Following the implementation and analysis carried out in Part 1, a discussion was held with the project supervisor to evaluate the results and determine the next steps.

A key takeaway from this discussion was that evaluating the enhancement method on only a few images is not sufficient to assess its effectiveness and robustness. The initial analysis was limited in scope, making it difficult to draw reliable and general conclusions.

To address this, it became necessary to extend the evaluation to a larger dataset in order to:

- Verify the consistency of the results
- Study the performance under varying lighting conditions
- Enable meaningful comparison between different enhancement approaches

This marked a shift from isolated image testing to a more comprehensive dataset-based evaluation.

B. Dataset Collection and Selection

To overcome the limitations of small-scale testing:

- A large dataset containing approximately 27,000 images was utilized
- The dataset primarily consists of night-time driving scenes captured from a moving vehicle

These images cover a variety of real-world conditions, including:

- Road environments
- Different street lighting conditions
- Dynamic outdoor scenarios

From the complete dataset:

- A subset of 50 low-light images was carefully selected
- These images were used for detailed experimentation and evaluation

Due to space limitations in the report:

- Only a few representative examples are presented
- However, all conclusions are derived from experiments conducted on the full set of 50 images

C. Experimental Objectives

The primary objectives of this phase are as follows:

- Compare the performance of different enhancement techniques across multiple images
- Evaluate the consistency and reliability of the results
- Analyze the effectiveness of adaptive versus fixed parameter approaches
- Improve the overall robustness and stability of the enhancement pipeline

D. Enhancement Strategies Explored

In this phase, three major enhancement approaches were implemented and compared.

1) **Adaptive Enhancement Approach (Mean-Median Based):** In this approach, the enhancement parameter is computed dynamically for each image based on its statistical properties.

$$\alpha = \frac{|\text{mean}(V) - \text{median}(V)|}{\text{mean}(V) + \text{median}(V)} \quad (2)$$

$$c_{\text{adaptive}} = \alpha \cdot \text{mean}(S) \quad (3)$$

Characteristics:

- Fully data-driven and image-specific
- Adapts according to the brightness distribution of the input
- Produces varying results depending on image characteristics

Good Cases:

- Extremely dark images, where strong enhancement significantly improves visibility
- Scenes with relatively uniform darkness, where adaptive scaling performs effectively

Bad Cases:

- Images with mixed lighting conditions, leading to over-enhancement in already bright regions
- Noisy images, where noise gets amplified along with brightness
- Images containing outliers, resulting in unstable parameter estimation

Example Analysis:

Example 1 (Good Case)



Fig. 2. Adaptive Enhancement — Effective Case (Low-Light Image)

Observation: A significant improvement in visibility is observed, with clearer structural details.

Example 2 (Bad Case)

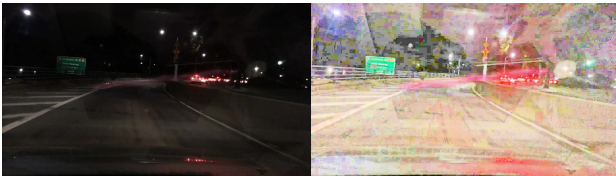


Fig. 3. Adaptive Enhancement — Over-Enhancement Case

Observation: Bright regions become saturated, resulting in an unnatural and over-enhanced appearance.

2) **Domain-Based Enhancement (Fixed Weight):** To address the instability of the adaptive method, a domain-based approach with a fixed parameter was introduced.

$$c_{\text{domain}} = w \cdot \text{mean}(S) \quad (4)$$

where $w = 3.68$.

Motivation:

- Reduce variability observed in the adaptive approach
- Achieve more consistent enhancement across different images

Characteristics:

- Uses a fixed parameter value
- Produces stable and predictable results across the dataset
- Less sensitive to noise and outliers

Good Cases:

- Images with moderate low-light conditions
- Scenes with relatively balanced brightness distribution

Bad Cases:

- Extremely dark images, where enhancement remains insufficient
- Images with already bright regions, where slight over-enhancement may occur

Example Analysis:

Example 3 (Stable Case)

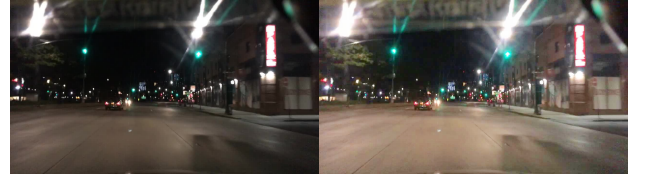


Fig. 4. Domain-Based Enhancement — Stable Output

Observation: The output shows consistent brightness with a more natural visual appearance.

Example 4 (Failed Case)

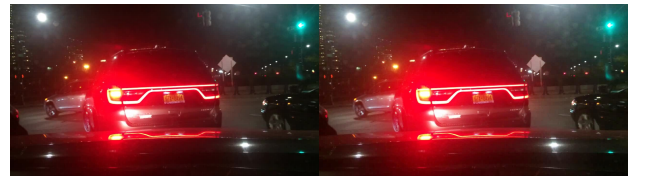


Fig. 5. Domain-Based Enhancement — Under-Enhancement Case

Observation: Very dark regions are not sufficiently enhanced, resulting in limited visibility improvement.

3) Updated Adaptive Enhancement (Darkness-Based):

To overcome the limitations observed in the original adaptive method, a modified approach based on image darkness was introduced.

$$\text{darkness} = 1 - \frac{\text{mean}(V)}{255} \quad (5)$$

$$c = \text{darkness} \cdot k \quad (6)$$

where $k \approx 50$.

Motivation:

- Establish a direct relationship between enhancement strength and overall image darkness
- Improve consistency and stability across different images

Characteristics:

- More stable compared to the original adaptive approach
- Provides better control over brightness levels

Good Cases:

- Performs well across a wide range of lighting conditions
- Suitable for both very dark and moderately illuminated images

Bad Cases:

- Performance is still limited in extreme lighting scenarios
- Requires careful tuning of the parameter k for optimal results

Example Analysis:

Example 5

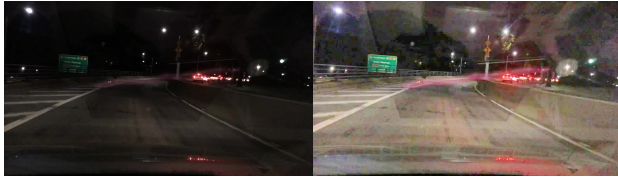


Fig. 6. Updated Adaptive Enhancement — Balanced Output

Observation: The output demonstrates a balanced improvement in brightness while preserving structural details.

E. Illumination Estimation Improvements

Two illumination estimation techniques were analyzed and compared:

Gaussian Filtering

- Simple and computationally efficient
- Tends to blur edges and reduce structural sharpness

Guided Filtering

- Preserves edges more effectively
- Produces sharper and more accurate illumination maps
- However, it has limited impact on the overall enhancement quality

F. Fusion Strategy Enhancements

To improve the quality of fusion and reduce artifacts observed in earlier methods, alternative fusion strategies were explored.

Variance-Based Fusion

- Coefficients are selected based on local variance, allowing regions with higher detail to be emphasized

Weighted Fusion

$$w_A = \frac{\text{var}_A}{\text{var}_A + \text{var}_B}, \quad w_B = \frac{\text{var}_B}{\text{var}_A + \text{var}_B} \quad (7)$$

$$C = w_A \cdot A + w_B \cdot B \quad (8)$$

Advantages:

- Produces smoother transitions between regions
- Reduces visible artifacts compared to max-based fusion

G. Evaluation Metrics

To quantitatively assess performance, the following metrics were used:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- AMBE (Absolute Mean Brightness Error)

H. Comparative Analysis

Method	Brightness	Stability	SSIM
Adaptive	High	Low	Moderate
Domain	Moderate	High	Stable
Updated Adaptive	Balanced	Medium-High	Improved

TABLE I
COMPARISON OF ENHANCEMENT METHODS

I. Key Insights

- Adaptive methods offer flexibility but suffer from instability across different images
- Fixed parameter methods provide consistent results but lack adaptability
- The updated adaptive approach achieves a better balance between flexibility and stability
- The design of the enhancement function plays a crucial role in overall performance

J. Final Conclusion of Part 2

The dataset-based evaluation highlights the strengths and limitations of different enhancement strategies. While adaptive approaches provide strong enhancement capability, they often lead to inconsistent results. In contrast, domain-based methods offer greater stability but are less responsive to varying image conditions. The updated adaptive method provides a more balanced solution, although the improvements remain incremental.

Overall, this phase establishes a solid experimental foundation for further refinement and optimization of the enhancement pipeline.

XII. PART 3: ADVANCED ENHANCEMENTS, MODEL REFINEMENT, AND APPLICATION-ORIENTED ANALYSIS

A. Motivation for Further Improvements

After completing the large-scale evaluation in Part 2, several observations were made:

- The improvements achieved through adaptive and domain-based tuning were relatively incremental
- Although many core issues were addressed, certain challenges still remained:
 - Slight over-brightness in some regions
 - Noise amplification under specific conditions
 - Loss of visual realism in certain scenarios
- There was clear potential to further improve the system for real-world applications

Based on a discussion with the project supervisor, a new direction was established:

Shift focus from parameter tuning to pipeline-level improvements and practical usability

B. Experimental Direction in Part 3

This phase was oriented towards enhancing both performance and applicability:

- Improving overall visual realism
- Minimizing artifacts and noise
- Effectively handling real-world night driving scenarios
- Designing a pipeline suitable for practical deployment

To achieve these objectives, three major pipelines were developed:

- Pipeline 1: Retinex + CLAHE + Denoising + White Balance
- Pipeline 2: Gamma-Based Enhancement
- Pipeline 3: Final Optimized HSV-Based Pipeline

C. Pipeline 1: Retinex-Based Enhancement with Adaptive and Domain Variants

1) **Motivation:** The Retinex-based approach was extended by incorporating both adaptive and domain-based parameter control. This allowed a comparative analysis of flexibility versus stability within the same framework.

2) Adaptive Variant:

- The parameter c was computed dynamically using image statistics
- The computation incorporates:
 - Mean–median relationship
 - Darkness-based scaling

Behavior:

- Produces strong enhancement in darker regions
- Exhibits high variability across different images

Good Case:

- Performs well for extremely dark scenes by significantly improving visibility

Bad Case:

- Leads to over-enhancement and saturation in already bright regions

3) Domain-Based Variant:

- A fixed parameter is used consistently across all input images
- The approach is designed to produce uniform and predictable results

Behavior:

- Provides stable and consistent enhancement
- Less aggressive compared to the adaptive variant

Good Case:

- Performs well for moderately dark images, producing balanced and visually pleasing results

Bad Case:

- Struggles with very dark images, often resulting in insufficient enhancement

4) Overall Observations:

- The adaptive approach offers flexibility but lacks consistency across different images
- The domain-based approach ensures stability but is less effective under extreme lighting conditions
- The Retinex-based pipeline, although powerful, is relatively complex and highly sensitive to parameter selection

5) Visual Results: Good vs Bad Cases: Good Case (Retinex + Adaptive)

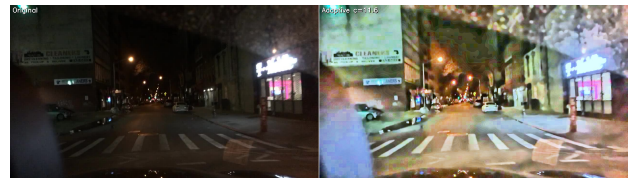


Fig. 7. Pipeline 1 (Adaptive) — Good Case: Enhanced visibility in dark regions

Explanation: The adaptive parameter effectively increases brightness in darker regions, making previously hidden details more visible while preserving structural information.

Bad Case (Retinex + Adaptive)

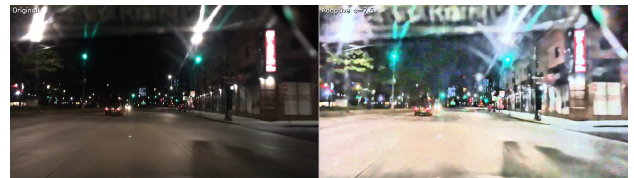


Fig. 8. Pipeline 1 (Adaptive) — Bad Case: Over-enhancement in bright regions

Explanation: Excessive enhancement causes bright regions to become saturated, resulting in loss of detail and an unnatural visual appearance.

Good Case (Retinex + Domain)

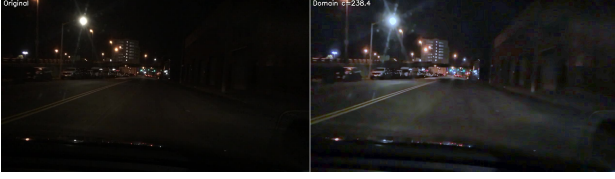


Fig. 9. Pipeline 1 (Domain) — Stable Enhancement

Explanation: The domain-based approach produces a balanced and consistent output, maintaining a natural appearance without excessive enhancement.

Bad Case (Retinex + Domain)



Fig. 10. Pipeline 1 (Domain) — Under-enhancement in very dark regions

Explanation: In extremely dark regions, the fixed parameter limits the level of enhancement, leading to insufficient brightness improvement.

D. Pipeline 2: Advanced Gamma-Based Enhancement with Adaptive and Fixed Control

1) **Motivation:** To reduce the complexity associated with Retinex-based methods explored in Pipeline 1, a gamma-based enhancement approach was developed. This method aims to control brightness more directly using a simpler transformation, while incorporating additional processing stages to improve contrast, suppress noise, and maintain color consistency.

Both adaptive and fixed gamma strategies were implemented to study the trade-off between flexibility and stability.

2) **Adaptive Gamma Formulation:** In the adaptive approach, the gamma value is computed dynamically based on the brightness of the image:

$$\gamma = \frac{\log(0.5)}{\log(\text{mean}(V))} \quad (9)$$

Constraints:

- $\gamma \in [0.5, 2.5]$

Interpretation:

- For darker images, a lower gamma value results in stronger enhancement

- For brighter images, the correction remains mild to avoid over-enhancement

3) **Fixed Gamma Approach:** In contrast, the fixed approach uses a constant gamma value:

$$\gamma = 1.2 \quad (10)$$

Purpose:

- Ensures consistent and predictable enhancement across all images
- Reduces instability introduced by adaptive parameter computation

4) **Complete Enhancement Pipeline:** The gamma-based enhancement pipeline consists of the following stages:

1. Color Space Conversion

- The image is converted from BGR to RGB, and then to HSV
- Enhancement is applied only to the Value (V) channel

2. Gamma Correction

$$V_{\text{gamma}} = V^{1/\gamma} \quad (11)$$

3. CLAHE (Contrast Enhancement)

- Clip limit: 2.0
- Tile size: 8×8
- Enhances local contrast while avoiding over-amplification

4. Denoising

- Non-local means filtering is applied
- Reduces noise before further enhancement stages

5. Controlled Gain Mechanism

$$\text{gain} = \frac{V_{\text{denoised}}}{V_{\text{original}}} \quad (12)$$

- The gain is constrained as:

$$\text{gain} \leq 3.0 \quad (13)$$

- The gain map is smoothed using Gaussian filtering

Role:

- Prevents excessive brightness amplification
- Ensures stable and controlled enhancement

6. Sharpening

- Unsharp masking is applied
- Enhances edges and structural details

7. Saturation Control

- Saturation is adjusted based on brightness changes
- Helps maintain natural color appearance and avoids distortion

8. Reconstruction

- The modified channels are merged and converted back to the BGR color space

5) **Visual Analysis of Results:** A comprehensive evaluation across multiple night driving scenes revealed the following observations:

Critical Observation: Adaptive Gamma Behavior

- The adaptive gamma approach did not produce consistently satisfactory results across the dataset
- Although visibility improved in some cases, it was often accompanied by significant over-enhancement and loss of detail

Conclusion:

The adaptive gamma approach is not a reliable method for low-light enhancement in real-world scenarios.

Fixed Gamma Behavior

- Produces stable and controlled enhancement across different images
- Prevents excessive saturation in bright regions
- Provides moderate and consistent brightness improvement

Performance in Very Dark Images

- Performs effectively in uniformly dark scenes
- Improves visibility without introducing significant artifacts
- Preserves structural details and avoids halo effects

Insight:

- The fixed gamma approach avoids aggressive amplification, making it more reliable under extreme low-light conditions

Limitations

- Provides limited enhancement in moderately dark images
- Lacks adaptability to varying illumination within a single scene
- Some finer details remain insufficiently enhanced

Conclusion:

- The fixed gamma method offers stability and performs well in very dark conditions
- However, it lacks adaptability and is not sufficient as a standalone solution

6) **Common Limitations:**

- Difficulty in achieving a balanced enhancement across both dark and bright regions
- The adaptive method suffers from instability and excessive enhancement
- The fixed method results in under-enhancement in several scenarios
- High sensitivity to strong light sources affects output quality
- Absence of explicit illumination modeling limits performance

7) **Example Illustration:** Since the adaptive gamma approach did not yield consistently reliable results, only a representative example of the fixed gamma method is presented.

Fixed Gamma (Effective Case in Very Dark Scenario)



Fig. 11. Fixed Gamma — Effective brightness enhancement in a very dark scene

Observation:

- Overall brightness is improved without excessive saturation
- Road visibility and structural details are enhanced
- Light sources remain controlled without prominent halo artifacts

Final Insight

Gamma-based enhancement alone is insufficient for handling complex real-world low-light scenarios.

While the fixed gamma approach provides stability and performs well in extremely dark conditions, the adaptive approach introduces instability and artifacts. This highlights the need for more structured pipelines that explicitly account for illumination and brightness variations.

E. Pipeline 3: Final Optimized HSV-Based Enhancement

1) **Motivation:** The Retinex-based approach was found to be complex and highly sensitive to parameter tuning, while gamma-based methods were comparatively simple but insufficient for handling diverse scenarios.

To address this, a balanced and more robust pipeline was developed.

2) **Key Components: HSV Processing**

- Enhancement is applied only to the V channel
- Hue (H) and Saturation (S) are preserved to maintain color consistency

Mild Gamma Correction

- Applies controlled brightness enhancement without over-amplification

CLAHE

- Enhances local contrast and improves visibility in darker regions

Denoising

- Applied prior to enhancement to suppress noise amplification

Controlled Gain

$$gain = \frac{V_{enhanced}}{V_{original}} \quad (14)$$

- Gain is limited using a predefined threshold
- The gain map is smoothed using Gaussian filtering to avoid abrupt transitions

Sharpening

- Restores fine structural details lost during processing

Saturation Compensation

- Adjusts saturation to prevent unnatural color shifts

3) Observations: Good Cases

- Produces the best overall visual quality among all tested approaches
- Achieves a good balance between brightness and contrast
- Minimizes artifacts and preserves natural appearance

Bad Cases

- Slight reduction in edge sharpness in some cases
- Not ideal for applications that require strong edge preservation

Example Placeholder

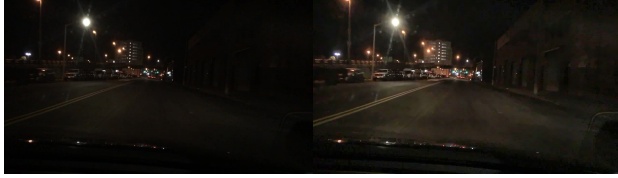


Fig. 12. Pipeline 3 Result (Final Model)

H. Application-Oriented Analysis

The focus was further extended towards practical applications, particularly in the domain of:

Night Driving Assistance

Objective:

- Improve overall road visibility under low-light conditions
- Enhance lane markings and structural edges for better interpretation

I. Key Insight

Best Visual Model $\neq$ Best Application Model

- Pipeline 3 performs best for human visual perception
- The updated adaptive method (from Part 2) is more effective for edge detection tasks

Reason:

- Pipeline 3 prioritizes smoothness and natural appearance, which can reduce edge sharpness
- Adaptive methods enhance contrast more aggressively, making edges more prominent

J. Final Insight

- More complex pipelines do not necessarily lead to better performance
- Effective brightness control plays a more critical role than precise illumination estimation
- The choice of enhancement method should be guided by the specific application requirements

F. Comparative Analysis of Pipelines

Feature	Pipeline 1	Pipeline 2	Pipeline 3
Brightness	High	Moderate	Balanced
Detail	Good	Moderate	Good
Noise Control	Good	Moderate	Good
Artifacts	Medium	Medium	Low
Stability	Low	Medium	High

TABLE II
COMPARISON OF ENHANCEMENT PIPELINES

G. Final Model Selection

Based on the comparative analysis, Pipeline 3 is selected as the most suitable approach for general use due to the following reasons:

- Produces visually natural and realistic results
- Preserves color fidelity effectively
- Minimizes artifacts across different scenarios
- Demonstrates consistent performance over the entire dataset

XIII. OVERALL CONCLUSION

This project addressed the challenge of low-light image enhancement, progressing from foundational understanding to advanced pipeline design and application-oriented evaluation.

Summary of Work

In Part 1, an existing research-based enhancement pipeline was studied and implemented. The method combined illumination estimation, adaptive brightness enhancement, and wavelet-based fusion. Although the approach improved overall brightness, it also introduced several issues such as over-enhancement, noise amplification, and inconsistent performance across different images. Further experimentation with guided filtering, modified fusion strategies, and contrast enhancement techniques indicated that the primary challenge lies in controlling brightness rather than accurately estimating illumination.

In Part 2, the proposed methods were evaluated on a larger dataset consisting of night driving scenes. Various enhancement strategies, including adaptive, domain-based, and updated adaptive approaches, were analyzed. The results showed that adaptive methods offer flexibility but often suffer from instability and over-enhancement, whereas domain-based methods provide stability but are less effective under challenging conditions. This phase emphasized the importance of achieving a balance between brightness enhancement and structural preservation.

In Part 3, the focus shifted towards developing improved pipelines and assessing their applicability in real-world scenarios. Three different pipelines were designed: a Retinex-based method, a gamma-based method, and a final optimized HSV-based pipeline. The analysis demonstrated that increased complexity does not necessarily result in better performance. While adaptive gamma introduced instability, fixed gamma offered stability but limited enhancement capability. The final optimized pipeline achieved the most effective balance between brightness, contrast, and visual realism.

Key Insights

The following key insights were derived from the study:

- Effective brightness control is more important than precise illumination estimation in low-light enhancement
- Adaptive methods provide flexibility but often lead to instability in real-world scenarios
- Fixed (domain-based) approaches ensure stability but lack adaptability across varying conditions
- Over-enhancement of bright regions remains a significant challenge in night-time images
- Combining multiple controlled processing steps yields better results than relying on a single transformation

Final Outcome

Among all the approaches explored, the final optimized HSV-based pipeline achieved the best overall performance in terms of visual quality, stability, and reduction of artifacts. However, it was also observed that the effectiveness of an enhancement method depends on the intended application. For instance, methods that enhance contrast more aggressively may be better suited for edge detection tasks, whereas balanced enhancement is preferable for human visual perception.

Future Scope

Future work can focus on the following directions:

- Development of learning-based models to improve adaptability and generalization
- Optimization for real-time implementation in practical driving systems
- Designing task-specific enhancement methods for applications such as edge detection and object recognition
- Improved handling of strong illumination sources, such as headlights and reflections

Final Remark

This project highlights that low-light image enhancement is not simply about increasing brightness, but involves maintaining a careful balance between illumination, contrast, noise, and perceptual quality. Effective solutions require thoughtful algorithm design along with controlled enhancement strategies to ensure consistent performance across diverse real-world conditions.

REFERENCES

- [1] P. Singh et al., "Low light image enhancement using reflection model and wavelet fusion," *Multimedia Tools and Applications*, Springer, 2024.